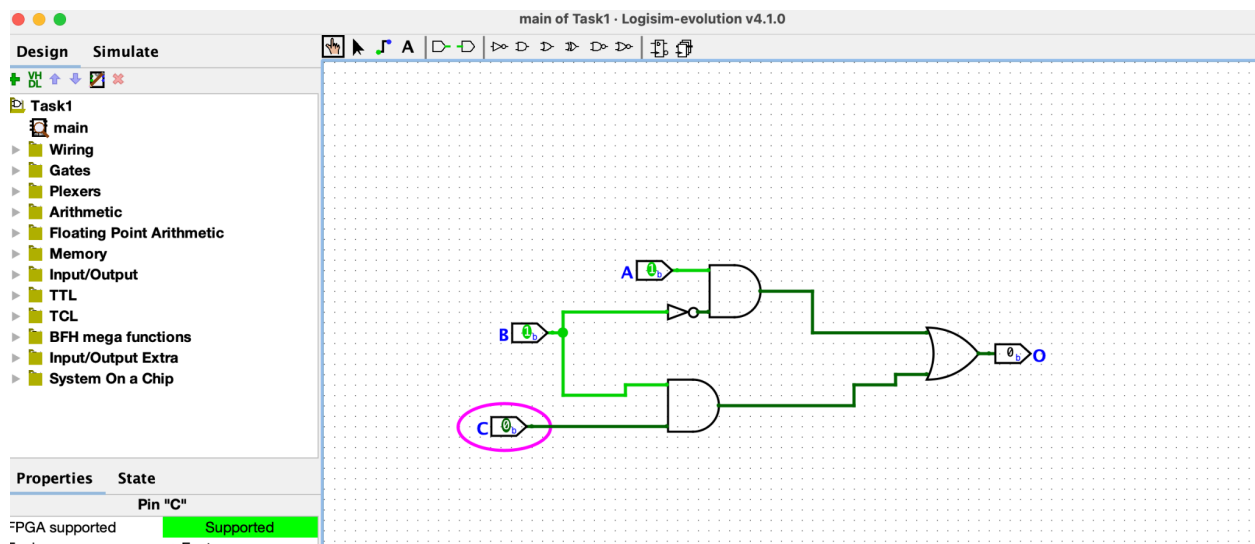


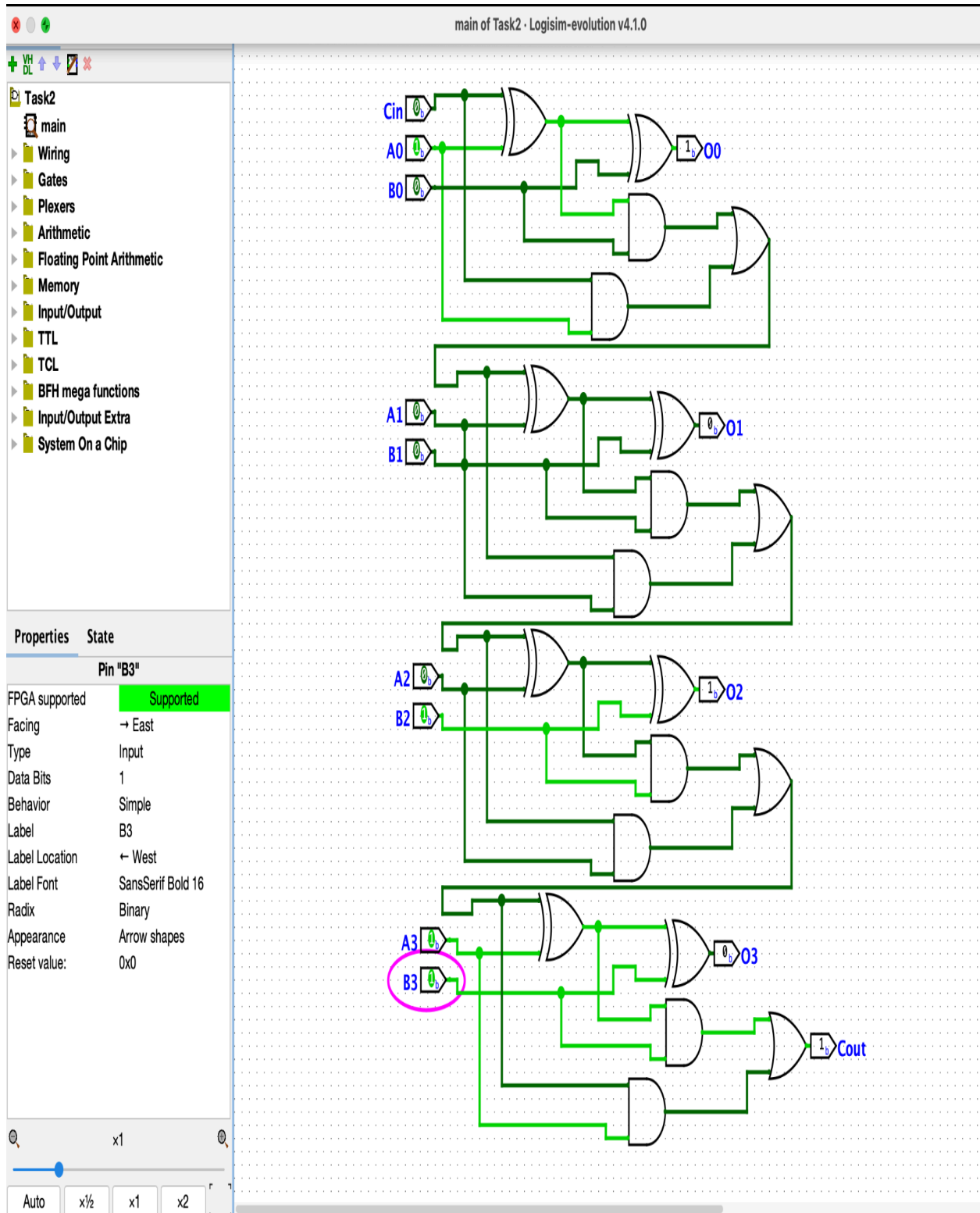
TASK 1

TASK 1 TRUTH TABLE

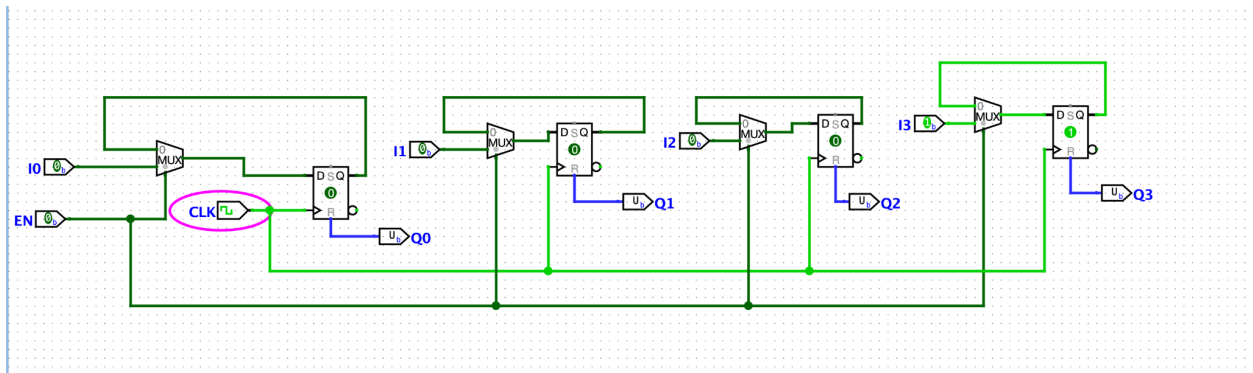
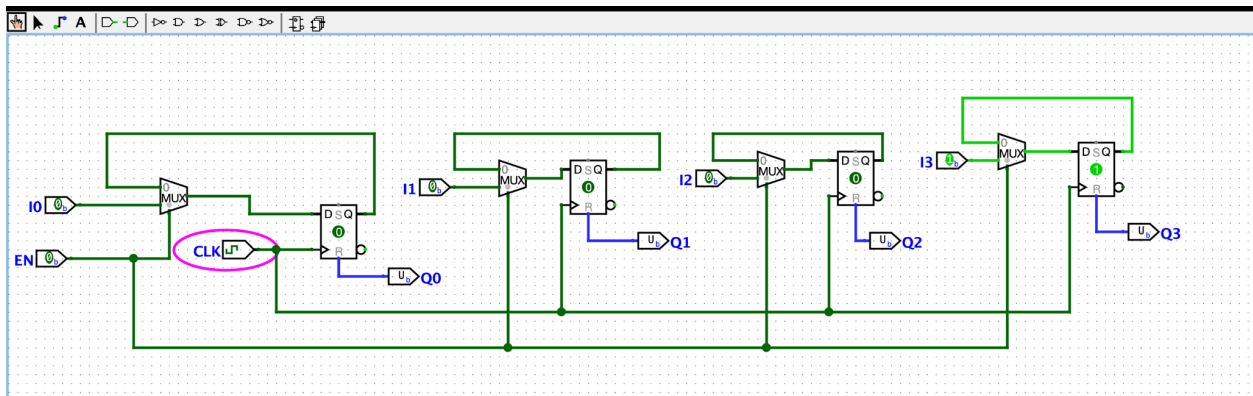
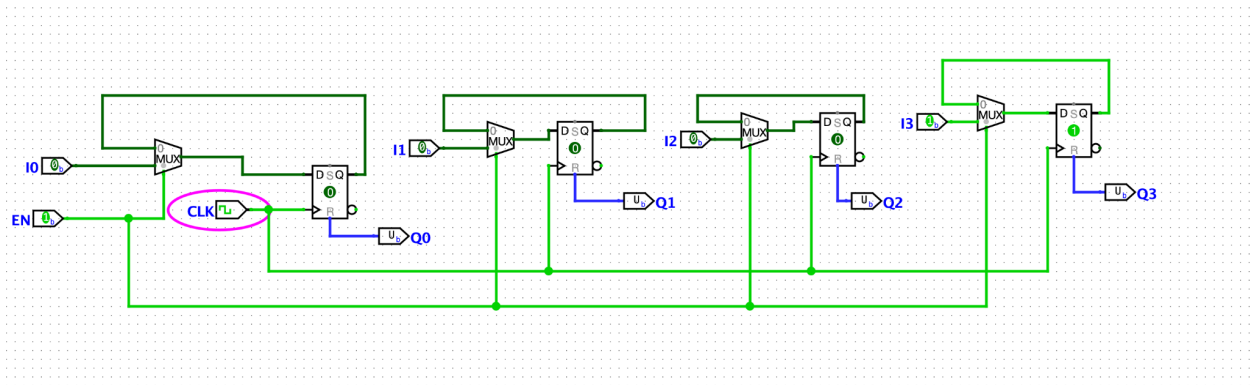
A	B	C	$\overline{B}$	$\overline{A}B$	BC	$\overline{A}\overline{B} + BC$
0	0	0	1	0	0	0
0	0	1	1	0	0	0
0	1	0	0	0	0	0
0	1	1	0	0	1	1
1	0	0	1	1	0	1
1	0	1	1	1	0	1
1	1	0	0	0	0	0
1	1	1	0	0	1	1



TASK 2



TASK 3



TASK 4

TASK 4

Derivation of equation

Since its 2 bit saturating counter implemented using D-flipflops

$\therefore 00 \rightarrow 01 \rightarrow 10 \rightarrow 11 \rightarrow 11 \rightarrow 11 \dots$
Saturation

input = Q_1, Q_0

output = D_1, D_0

$Q_1, Q_0 \rightarrow$ Present state

$D_1, D_0 \rightarrow$ Next state

Inputs

Q_1, Q_0
0 0
0 1
1 0
1 1
...

Outputs

D_1, D_0
0 1
1 0
1 1
1 1
...

continuation in end
as it is saturating

Truth table

Q_1	Q_0	D_1	D_0
0	0	0	1
0	1	1	0
1	0	1	1
1	1	1	1

Deriving equation for D_0 using K-map

$Q_0 \backslash Q_1$	0	1
0	1	1
1	0	1

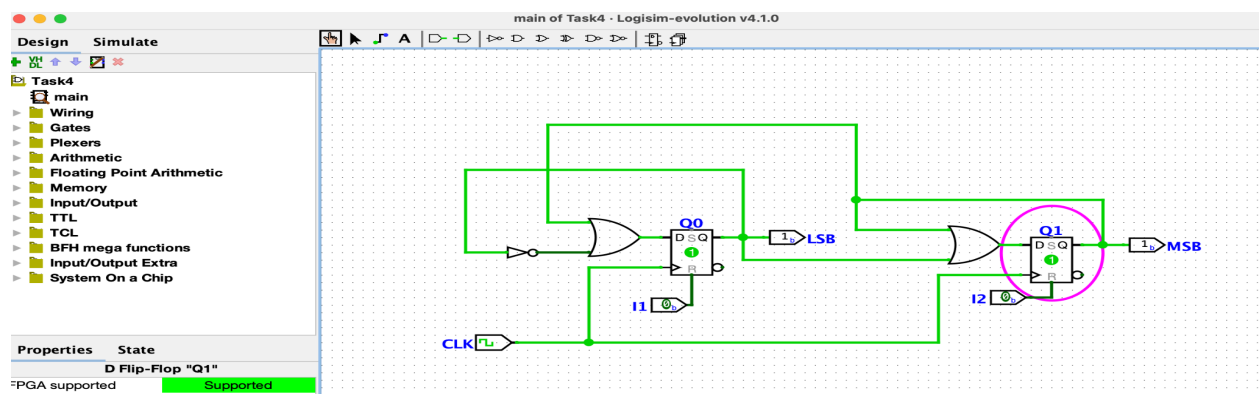
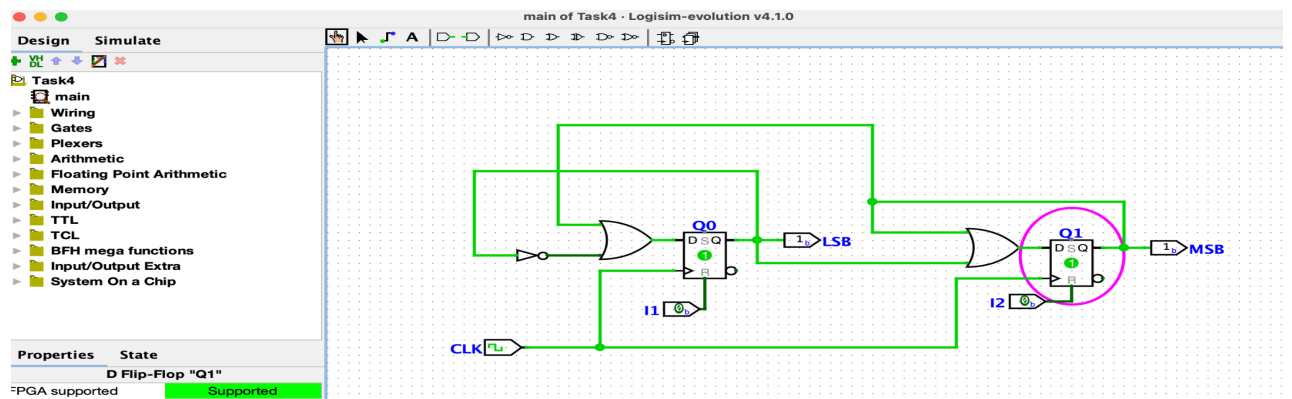
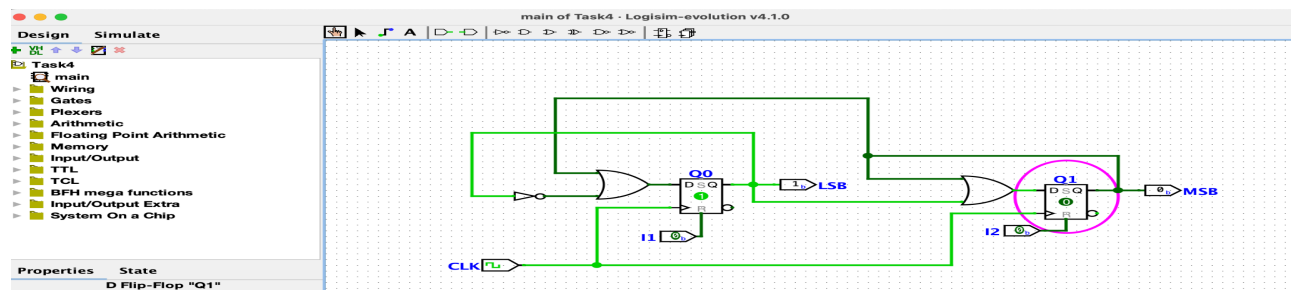
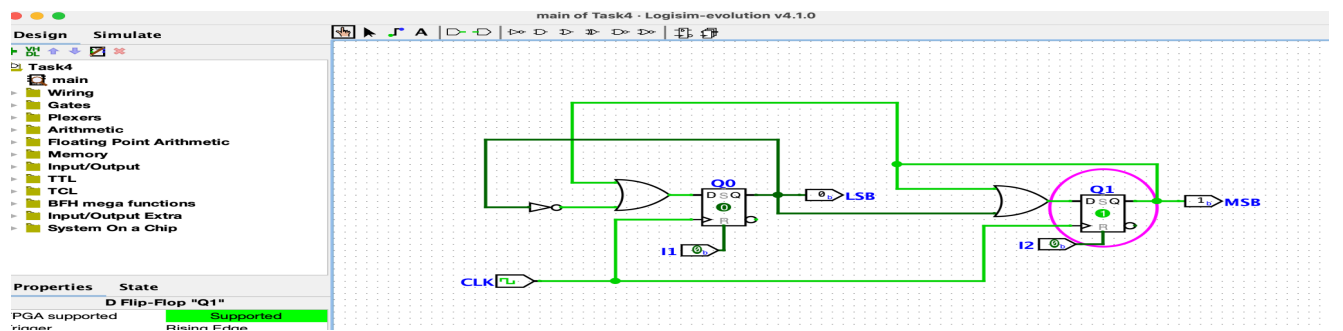
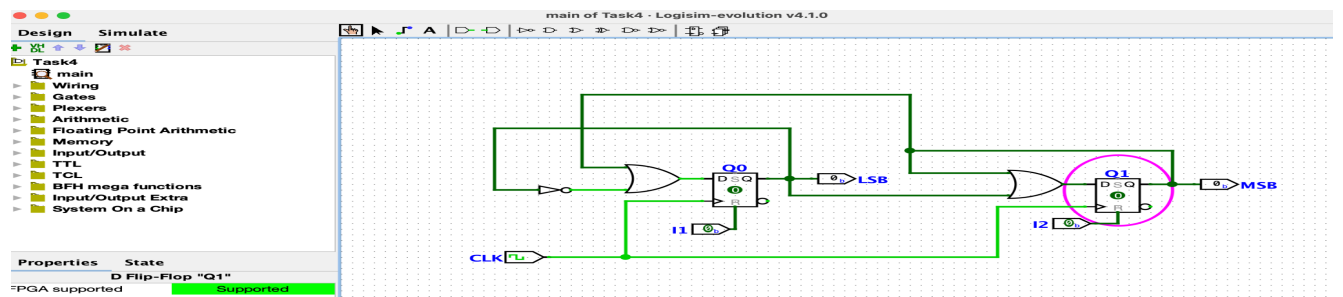
$$D_0 = \bar{Q}_0 + Q_1$$

Prime
Next
state
equation

For D_1

$Q_0 \backslash Q_1$	0	1
0	0	1
1	1	1

$$D_1 = Q_1 + Q_0$$



TASK 4 UPDATED

Task 4 updated

With both up and down counter.

Let us have U for going to either up or down counter.

$U=1 \rightarrow$ increment & $U=0 \rightarrow$ decrement

Q_1	Q_0	$Q_1(\text{next})$	$Q_0(\text{next})$	U
0	0	0	0	1
0	1	0	0	1
1	0	1	1	1
1	1	1	1	1
0	0	0	0	0
0	1	0	0	0
1	0	0	1	0
1	1	1	0	0

Using K-maps

$Q_0(\text{next})$

$Q_1 \backslash Q_0$	0	1
0	0	1
1	0	1

$\therefore Q_0(\text{next})$

$$= Q_1 \bar{Q}_0 + Q_1 + U(\bar{Q}_1 \bar{Q}_0)$$

$$\therefore Q_0(\text{next}) = Q_1(\bar{Q}_0 + 1) + U\bar{Q}_1\bar{Q}_0$$

$$Q_0(\text{next}) = Q_1(\bar{Q}_0 + U) + U(\bar{Q}_1 \bar{Q}_0)$$

Similarly, for $Q_1(\text{next})$

$Q_1 \backslash U$	0	1
00	0	0
01	0	1
10	1	1
11	0	1

$\therefore Q_1(\text{next})$

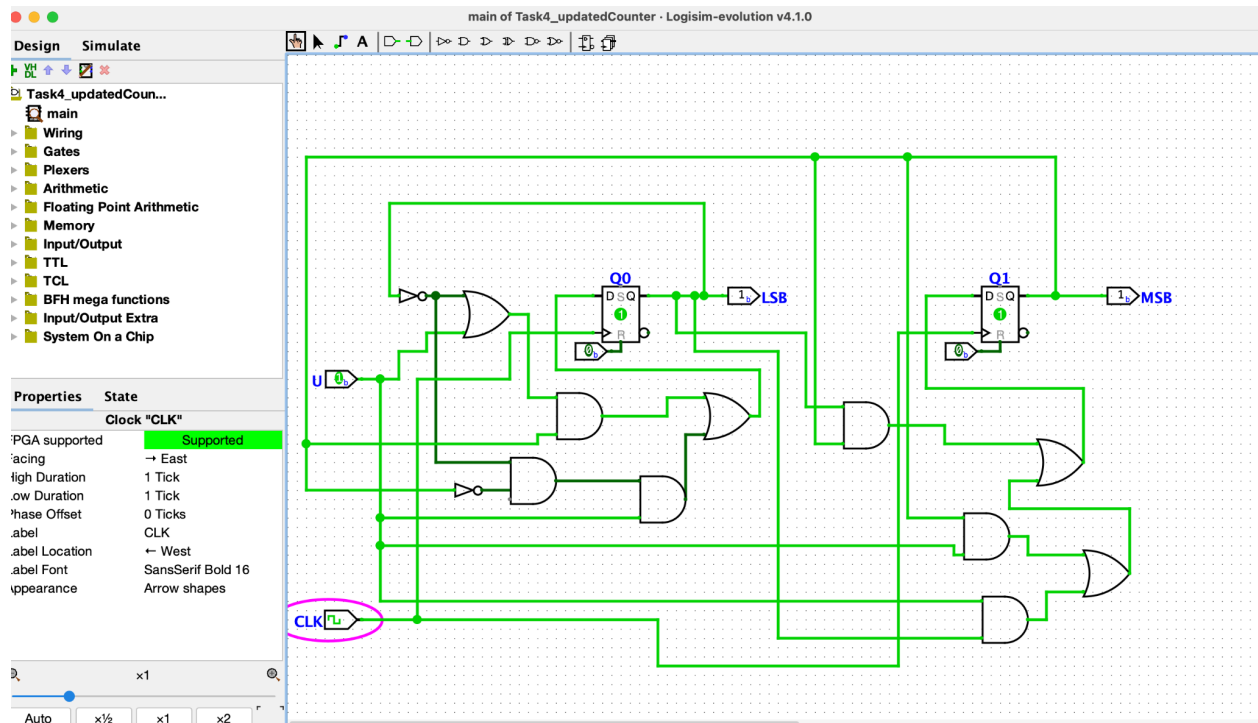
$$= Q_1 Q_0 + Q_1 U + Q_0 U$$

~~Sum~~

$$Q_0(\text{next}) = Q_1(\bar{Q}_0 + U) + U(\bar{Q}_1 \bar{Q}_0)$$

$$Q_1(\text{next}) = Q_1 Q_0 + Q_1 U + Q_0 U$$

hence, derived



Screenshot of 11 being saturated at the end of 5 cycles

(Answers of task 4 updated with both up and down counter)

Nandini Jain
CS24B035